

Copyright

What Can Be Protected

Copyright covers original expressions fixed in a tangible medium, but not ideas, facts, or short phrases.

- Books (novels, poems, articles), music (compositions, lyrics), films (movies, videos), software (code, programs), websites (design, content).
- Also includes databases, choreography, paintings, sculptures, architecture, maps, and advertisements.

Protection arises automatically upon creation; registration strengthens legal enforcement.

Rights of Copyright Owner

Owners hold a bundle of exclusive rights, often grouped into reproduction, distribution, and communication to the public.

- **Reproduction:** Copying the work in any form (print, digital, recordings).
- **Distribution:** Selling, renting, lending, or transferring copies to the public (physical or digital, subject to first-sale limits).
- **Communication to the Public:** Performing, displaying, broadcasting, or streaming publicly (includes digital transmissions for sound recordings).

Additional rights cover derivatives (adaptations) and moral rights (attribution, integrity).

Owners can license, assign, or enforce via lawsuits

How long does copyright protection last

Copyright protection duration varies by jurisdiction, creation date, authorship type, and whether the work is published, but follows international standards like the Berne Convention (life plus at least 0 years).

US Copyright Durations

Under US law (post-1978 works):

- Individual authors: Life of the author plus 70 years.

- Joint works: Life of the last surviving author plus 70 years.
- Works for hire, anonymous, or pseudonymous: 95 years from publication or 120 years from creation, whichever is shorter.

Pre-1978 rules are complex, often up to 95 years from publication if renewed.

India Copyright Durations

- Literary, dramatic, musical, artistic works: Author's life plus 60 years.
 - Cinematograph films, sound recordings: 60 years from publication.
- Anonymous/pseudonymous: 60 years from publication, shifting to life-plus if author revealed.

Key Notes

Terms end on December 31 of the expiration year; works then enter the public domain. EU and many countries match US life-plus-70; always check local laws for specific.

What are fair use exceptions to copyright

Fair use provides limited exceptions to copyright infringement, allowing certain unauthorized uses of protected works without permission to promote free expression, education, and innovation. It balances creators' rights with public needs, though rules differ by country, US fair use is flexible, while India's "fair dealing" is narrower.

US Fair Use Factors

US courts weigh four statutory factors from Section 107 of the Copyright Act.

- Purpose and character (e.g., transformative, non-commercial, educational uses favored).
- Nature of the work (factual vs. creative).
- Amount used (small portions preferred).
- Market effect (no significant harm to original's sales or value).

Common Fair Use Examples

Permitted scenarios include criticism, parody, news reporting, teaching, scholarship, and research.

- Quoting book excerpts in a review or academic paper.
- Parody videos mocking a song.
- Short clips in news for current events.
- Classroom handouts of brief articles.

India Fair Dealing (Section 52)

India lacks broad "fair use"; instead, specific fair dealing exceptions apply.

- Private/personal use, research, criticism/review, current events reporting.
- Public recitations of reasonable extracts.
- Judicial proceedings or legislative use.

Transformative works may qualify if adding value, but courts decide case-by-case.

Fair use is not a blanket right, it's judged holistically, often in court, and does not cover adaptive/derivative works without permission.

Digital Copyright

Copyright in Digital Environment

Copyright in the digital environment extends traditional protections to online activities, treating them as exercises of core exclusive rights like reproduction and communication to the public.

Uploading and Downloading as Reproduction

Uploading a copyrighted file to a server or cloud creates a copy, infringing the reproduction right, while downloading saves another copy to your device, both unauthorized acts unless permitted (e.g., fair use or license). Peer-to-peer sharing amplifies this, as uploaders distribute copies en masse.

Streaming as Communication to the Public

Streaming transmits content to users simultaneously, equating to "communication to the public" under laws like India's Copyright Act Section 37 (post-2012 amendments covering internet broadcasts). Platforms must obtain licenses or statutory permissions, paying royalties;

unauthorized live streams or embeds (e.g., YouTube without rights) infringe, even without downloads.

Temporary Storage and Cache Issues

Browser caches, RAM copies during viewing, or transient buffers create temporary reproductions, often exempted if incidental and non-persistent (e.g., US DMCA safe harbors, EU InfoSoc Directive Article 5(1), India's Section 52(1)(b)). However, deliberate caching for later access or server-side storage without rights risks infringement; courts distinguish "volatile" tech copies (permitted) from user-controlled ones (prohibited).

Technological Protection Mechanism

DRM(digital rights management)

Digital Rights Management (DRM) is a set of technologies and processes designed to control access to and usage of copyrighted digital content, such as software, music, videos, eBooks, and documents. It aims to protect creators' rights by preventing unauthorized copying, sharing, or modification.

How DRM Works

DRM creates a protective layer around digital files through encryption, which scrambles content so it can only be accessed with a valid key or license. Key components include authentication (verifying user identity and device), license management (defining permissions like time limits or device restrictions), and enforcement (blocking actions like printing or screenshots). Metadata embedded in files tracks ownership and usage rules, while features like watermarks or IP-based locks add further controls.

Key Features

- Prevents saving, editing, sharing, or screen-capturing protected content.
- Limits access by device count, location, expiration date, or usage frequency.
- Supports business models like subscriptions, rentals, or geo-restricted releases.

Common Use Cases

Publishers use DRM for eBooks and articles to enforce device limits and copyright compliance. In corporate settings, it safeguards sensitive data with view-only permissions and expiration rules. Media streaming services rely on it for anti-piracy in videos, often via standards like AES-128 encryption.

Benefits and Drawbacks

DRM reduces piracy and revenue loss for creators while enabling flexible licensing. However, it can frustrate legitimate users with restrictions and has faced criticism for limiting fair use or interoperability. Modern systems integrate with digital asset management (DAM) tools for automated compliance.

Common Use Cases

DRM spans entertainment, education, and enterprise.

- Streaming platforms protect movies/TV shows.
- Software verifies licenses to prevent cracks.
- eBooks limit copying/printing.
- Corporate files restrict forwarding or editing.

Major DRM Technologies

These systems encrypt content and enforce access rules via licenses.

- **Apple FairPlay**: Encrypts media for iOS, macOS, and Safari; used in iTunes, Apple TV+, and apps to block unauthorized playback.
- **Google Widevine**: Supports Android, Chrome, Roku, and smart TVs with levels of security.
- **Microsoft PlayReady**: Secures Windows software, Xbox games, and streaming; requires activation keys for installs
- Other examples include Adobe Access for PDFs/eBooks and HDCP for HDMI video outputs.

When you buy a movie on iTunes, FairPlay DRM ties it to your Apple ID and approved devices, preventing shares outside your account. Netflix uses Widevine on Android TVs to encrypt shows, stopping screen recordings and ensuring only your login streams content, even if someone guesses your password. Installing Microsoft Office prompts a product key check via PlayReady, blocking pirates while letting legit users activate on one PC. For everyday files, Adobe DRM on shared PDFs lets you view but not print or forward sensitive reports from work.

DRM and TPMs

Technical Protection Measures (encryption, access controls) are legally shielded.

- Anti-circumvention laws (US DMCA §1201, EU InfoSoc Art.6, India §65A) prohibit bypassing TPMs even for fair use; 3-year triennial rulemaking allows narrow exemptions (e.g., jailbreaking phones).
- Right holders sue for trafficking circumvention tools (e.g., DeCSS for DVDs); violations carry criminal penalties. TPMs enable licensing but spark debates on overreach.

Business Models in Digital Content

Licensing vs ownership (Spotify/Netflix model)

Licensing and ownership represent two distinct models for accessing digital content, especially in streaming services like Spotify and Netflix. In the licensing (subscription) model, users pay recurring fees for temporary access to a vast library without owning the content, while ownership grants permanent rights to specific items but lacks the breadth of a subscription catalog.

Licensing Model (Spotify/Netflix)

Users subscribe monthly for access rights, not possession.

- Content rotates based on deals; tracks or shows can vanish if licenses expire.
- Spotify licenses nearly all music from labels, paying royalties per stream; Netflix mixes licensed hits with owned originals.
- Benefits include low upfront costs, endless variety, and updates; drawbacks are "subscription fatigue" and no permanence.

Ownership Model

Users buy individual items for indefinite personal use.

- Purchases (e.g., iTunes songs, Amazon movies) stay in your library forever, playable offline without fees
- No library-wide access; you select specifics, avoiding rotation risks
- Suited for favorites but costlier for broad consumption.

Spotify vs. Netflix Comparison

Aspect	Spotify (Heavy Licensing)	Netflix (Mixed Ownership/Licensing)
Content Control	Licenses from labels; high royalties eat margins ☹☹☹	Owns originals (e.g., \$17B/year investment) for stable costs
User Experience	Access millions of tracks; lose songs if deals end ☹	Core library stable; licensed shows rotate
Economics	Revenue-tied payouts limit profits ☹	Ownership boosts long-term margins
Relatable Example	Your playlist song disappears post-license; re-subscribe to regain ☹	"Stranger Things" always there as Netflix-owned

This shift favors access for convenience but erodes traditional ownership, sparking debates on long-term value.

Interface with Copyright Law

Copyright law interfaces intensely with digital technologies, driving most modern litigation due to easy copying, distribution, and access online. Courts worldwide balance innovation with protection, often via safe harbors and anti-circumvention rules.

Framing and Linking

Framing embeds another site's content in your page's frame (e.g., inline display), while deep linking bypasses homepages to sub-content.

- Framing often infringes display rights if it implies endorsement or harms the original's market; US cases like *Perfect 10 v. Amazon* ruled non-infringing if not altering content.

- Linking is generally legal (no reproduction), but inline linking to thumbnails can violate if substituting the original—e.g., EU's Svensson case permits if not new public communication. Deep links evade liability unless facilitating infringement.

Caching Legality

ISPs and sites cache web pages for speed, creating temporary copies.

- Exempt if automated, non-alterative, and user-accessible only (US DMCA §512(b), EU Directive 2001/29/EC); retention beyond need risks infringement.
- India Copyright Act §52(1)(b) allows transient storage for transmission; deliberate caching (e.g., Google Cache) requires "volitional conduct" test for liability.

Software Piracy Protections

Software qualifies as "literary works," protecting source/binary code.

- Unauthorized copying, decompiling (except interoperability), or cracking licenses infringes reproduction/derivative rights; WIPO Copyright Treaty mandates this.
- EULAs enforce single-user limits; reverse engineering permitted narrowly (e.g., US 17 USC §117 for backups).

Intermediary Liability and Safe Harbors

Platforms like YouTube/ISPs avoid liability for user content if passive.

- **US DMCA §512:** Notice-and-takedown—register agent, remove flagged content expeditiously, no "red flag" knowledge; repeat infringers terminated.
- **India IT Act §79 + Copyright Act §52(1)(c):** Similar immunity if acting as conduit, no abetment, expeditious takedown post-notice.
Repeat offenders lose protection; proactive filtering not required.

Common Cyber Issues

Common cyber issues like piracy and plagiarism intersect with copyright law, posing legal, academic, and security risks in the digital age. Torrent sites and illegal streams drive massive infringement, while distinctions like plagiarism versus infringement clarify accountability.

Piracy via Torrent Websites

Torrents use peer-to-peer (P2P) networks where users simultaneously upload (seed) and download copyrighted files like movies or software, infringing reproduction and distribution rights. Sites like The Pirate Bay facilitate this, exposing users to lawsuits (e.g., \$1M+ damages in US cases) and malware—studies show pirate sites carry 20-65% higher malware risks than legit ones, including ransomware and credential theft.

Illegal Streaming

Unauthorized streams violate communication-to-public rights, as servers transmit content without licenses. Rogue sites (e.g., FMovies clones) embed malware in ads/pop-ups; Interpol links them to organized crime, with doubled exposure time raising malware hits by 20%.

Plagiarism vs. Copyright Infringement

Aspect	Plagiarism	Copyright Infringement
Nature	Ethical/academic breach: Failing to attribute/credit sources	Legal violation: Unauthorized reproduction/distribution/display
Targets	Offense against author (reputation harm)	Offense against copyright holder (often publisher via CTA)
Enforcement	Honor codes, school discipline, journal retractions	Federal courts, damages, injunctions
Overlap	Possible (e.g., copied text without quotes) but distinct—plagiarism of public domain or fair use isn't infringement	Can occur without plagiarism if attributed but unlicensed

Academic difference: Plagiarism violates integrity norms (e.g., no citation in essays); legal copyright focuses on economic rights, common even with credit.

Fair Use/Fair Dealing in Education, Research, Criticism

US fair use flexibly permits "short excerpts" for teaching/research/criticism if transformative and non-market harming (e.g., quoting in lectures). India's fair dealing (Copyright Act §52) is narrower, allowing specific uses like private study or reviews but not broad copying—guidelines limit to 10% excerpts. Both protect parody/news but require balancing; educators often qualify, though DMCA exemptions aid circumvention for fair purposes.

Software Copyright

Software copyright treats computer programs as "literary works," protecting their expression but not underlying ideas, algorithms, or functionality.

What Gets Protected

- **Source Code:** Human-readable instructions (e.g., Python, Java scripts) qualify for full copyright upon original creation, covering structure, sequence, and organization.
- **Object Code:** Compiled machine-readable binaries are also protected, as copying either form infringes reproduction rights.

Exclusions: Functional elements like menus or methods of operation; interfaces may get limited visual protection.

Licensing Models

Licenses dictate usage terms despite underlying copyright.

Open-Source Licensing

Source code is publicly available, promoting collaboration.

- **MIT License:** Permissive—allows free use, modification, distribution (even in proprietary products); requires only license/copyright notice retention.
- **GPL (GNU General Public License):** Copyleft—requires derivatives to stay open-source under GPL, mandating source code sharing; prevents proprietary lock-in.

License	Modification Allowed?	Distribution Requirements	Commercial Use
MIT	Yes	Include notice; no source share needed	Yes, even closed-source
GPL	Yes	Share source; derivatives GPL	Yes, but must open modifications

Proprietary Software

Source/object code kept secret; users get restricted licenses via EULAs.

- Single/multi-user limits, no reverse-engineering (except narrow interoperability exceptions).
- Examples: Microsoft Windows, Adobe apps—cracking violates anti-circumvention laws like DMCA.

Open-source fosters innovation via community; proprietary prioritizes control and revenue.

Important Legal Mechanisms

Takedown notices and intermediary liability rules are key mechanisms shielding platforms from copyright lawsuits while empowering rights holders to combat infringement efficiently. These systems, inspired by the US DMCA, have global analogs, including India's IT Act and Copyright Act provisions.

Takedown Notices (DMCA-Style)

Copyright owners send formal notices to platforms hosting infringing content, triggering swift removal without court orders.

- **Process:** Include signature, contact info, infringing URL, original work description, good-faith infringement statement, and agent info; platforms must remove "expeditiously."
- **Counter-Notice:** Accused users reply under penalty of perjury claiming error; platforms restore after 10-14 days unless sued.
- Platforms like YouTube use automated forms; abuse risks perjury penalties or countersuits.

Intermediary Liability

Passive hosts (YouTube, Google, ISPs) gain "safe harbor" immunity if they lack direct knowledge/control and respond to notices.

- **US DMCA §512:** No liability for user content if designated agent, no financial benefit from infringement with takedown ability, and terminate repeat offenders.

- **India IT Act §79 + Copyright Act §52(1)(c):** Immunity as "intermediaries" requires due diligence, no active facilitation, expeditious takedown post-notice; 2021 IT Rules mandate grievance officers.
- Cases like *Shree Krishna v. Google* test boundaries—courts deny safe harbor if platforms ignore "red flags" or fail notice protocols.

Mechanism	US (DMCA)	India (IT Act)
Immunity Conditions	Passive host, agent registered, swift removal	Due diligence, no abetment, takedown post-notice
Notice Requirements	Specific URLs, ownership proof	Formal complaint with evidence
Platform Duties	Forward counters, wait 0- days	Grievance officer, user termination for repeats

These balance enforcement with free speech; non-compliance exposes platforms to damages.